46fbc41d-0a2e-46b7-870d-e90563e05597

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'BlackbirdWorks/gopherstack' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gopherstack/gopherstack (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-13T16:09:12.406Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Implement SageMaker Runtime service
</pr_title>

<pr_description>
- [x] Create `services/sagemakerrumtime/` package (backend.go, handler.go, provider.go, handler_test.go)
- [x] Create dashboard handler `dashboard/sagemakerrumtime_handlers.go`
- [x] Create dashboard template `dashboard/templates/sagemakerrumtime/index.html`
- [x] Update `dashboard/ui.go` (import, SageMakerRuntimeOps fields, template pattern, route setup, nav tab)
- [x] Update `dashboard/provider.go` (interface method,

extraction logic, extractedConfig field, config assignment)
- [x] Update `dashboard/doc_handlers.go` (SageMakerRuntimeOps field)
- [x] Update `dashboard/provider_test.go` (mock method)
- [x] Update `cli.go` (import, provider registration, handler getter)
- [x] Update `internal/teststack/teststack.go` (import, handler field, initialization, registration, Stack exposure)
- [x] Create `test/e2e/sagemakerrumtime_test.go`
- [x] **Fix RouteMatcher comment** - remove incorrect "POST requests" claim; matcher only checks path + SigV4 service name
- [x] **Fix InvokeEndpointWithResponseStream** - use proper AWS binary event stream framing (same as BedrockRuntime)
- [x] **Fix Go snippet** - removed unused `strings` import; use `io.ReadAll` to read response body
- [x] **Add SageMaker Runtime to doc.html** - SageMakerRuntimeOps operations now rendered in Docs page
- [x] 93.9% test coverage; lint passes with 0 issues; build succeeds
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

hor>
copilot-autodev: This PR is addressing Issue #458.
</comment_old>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Pull request overview

Adds a new **SageMaker Runtime** service implementation (`sagemaker-runtime`) to the emulator, including a minimal REST handler + in-memory backend, dashboard UI for invocation visibility, and wiring into CLI/teststack/E2E.

Changes:
- Introduces `services/sagemakerrumtime/` with handler/backing store/provider + unit tests.
- Adds a dashboard page + template to display recorded runtime invocations and usage snippets.
- Wires the new service through CLI, dashboard provider/config, teststack, and adds an E2E UI test.

Reviewed changes

Copilot reviewed 13 out of 13 changed files in this pull request and generated 4 comments.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>dashboard/sagemakerrumtime_handlers.go:36-46</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
The Go snippet appears to include an unused import (`strings`) and doesn't show the typical pattern for reading the InvokeEndpoint response body. Since snippets are intended to be copy/pasteable, consider removing unused imports and showing reading from the response body (e.g., via io.ReadAll) before logging/printing.
</comment_old>

</comment_thread>

<comment_thread>
<file>services/sagemakerrumtime/handler.go:172-186</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
InvokeEndpointWithResponseStream responses use the AWS event stream binary framing; returning a raw JSON blob here (even with Content-Type set to application/vnd.amazon.eventstream) will cause AWS SDK clients to fail decoding the stream. Consider encoding a valid event stream frame (similar to services/bedrockruntime's encodeEventStreamMsg) and writing that instead of plain JSON bytes.

</comment_old>

</comment_thread>

<comment_thread>
<file>services/sagemakerrumtime/handler.go:65-78</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
RouteMatcher's comment says it matches POST requests, but the matcher currently only checks path prefix and SigV4 service name (no method check). Either update the matcher to also require POST, or adjust the comment so it matches the actual routing behavior.
</comment_old>

</comment_thread>

<comment_thread>
<file>dashboard/doc_handlers.go:153-157</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
docPageData now includes SageMakerRuntimeOps and docIndex populates it, but dashboard/templates/doc.html currently only renders .SageMakerOps (no SageMaker Runtime section). As-is, SageMaker Runtime operations won't appear in the Docs page despite being computed here; consider updating the doc template to display SageMakerRuntimeOps.
</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@agbishop</author>
@copilot (refinement check 1 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please address any review feedback.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot (refinement check 1 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please addres...

Addressed all four review items in commit `65d1ab3`:

1. **RouteMatcher comment** — updated to accurately describe the matching behavior (path prefix + SigV4 service name check; no POST-only restriction)
2. **InvokeEndpointWithResponseStream** — now returns a proper AWS binary event stream frame (same `encodeEventStreamMsg` framing as BedrockRuntime) instead of raw JSON
3. **Go snippet** — removed the unused `strings` import and replaced `out.Body` logging with `io.ReadAll` to correctly read the response stream
4. **doc.html** — added a SageMaker Runtime section so `SageMakerRuntimeOps` operations are visible on the Docs page
</comment_old>

<comment_new>
<comment_id>4056155475</comment_id>
<author>@agbishop</author>
copilot-autodev: PR requires manual deployment approval for workflow(s): Pull Request CI. Waiting for a reviewer to approve the environment deployment before proceeding.
</comment_new>

<comment_new>
<comment_id>3945215736</comment_id>
<author>@agbishop</author>
@copilot (refinement check 2 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please address any review feedback.
</comment_new>

</pr_comments>

</comments>

The last **4** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement><rules_for_making_changes>

* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

<prev_assistant>

Here's a summary of what was done in refinement check 2 (commit `dc0b77a`):

Gaps found and fixed:

1. **Go snippet package name** — the snippet was referencing `sagemakerrumtime` (a non-existent package). The actual AWS SDK Go v2 package is `github.com/aws/aws-sdk-go-v2/service/sagemakerruntime`. Also corrected the response reading: `InvokeEndpointOutput.Body` is `[]byte` (not `io.ReadCloser`), so `io.ReadAll` was removed in favor of direct use.

2. **Missing `dashboard/sagemakerrumtime_handlers_test.go`** — created with 3 parallel table-driven test cases covering: page header render, recorded invocation render, and empty state message.

3. **Missing Terraform fixture** (`test/terraform/fixtures/sagemakerrumtime/success.tf`) — the issue explicitly requires Terraform test fixtures. Created following the BedrockRuntime pattern (comment-only fixture, SDK-direct verification in the test).

4. **Missing `TestTerraform_SageMakerRuntime`** — added to `terraform_test.go` with the `createSageMakerRuntimeClient` helper in `main_test.go` (using the correctly-named `sagemakerruntime` SDK). The test verifies `InvokeEndpoint` returns a non-empty response body.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'BlackbirdWorks/gopherstack' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gopherstack/gopherstack (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-13T16:51:30.055Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Implement SageMaker Runtime service
</pr_title>

<pr_description>
Adds the missing SageMaker Runtime service (`sagemaker-runtime`) — the inference invocation API distinct from the SageMaker management service.

Service (`services/sagemakerrumtime/`)
- REST handler matching `/endpoints/{EndpointName}/...` gated on SigV4 service name `sagemaker-runtime` (priority 85)
- Three operations: `InvokeEndpoint`, `InvokeEndpointAsync`, `InvokeEndpointWithResponseStream`
- `InvokeEndpointWithResponseStream` uses proper AWS binary event stream framing (same protocol as BedrockRuntime)

```
- In-memory backend recording invocations for dashboard
visibility

```bash
# CLI usage
aws sagemaker-runtime invoke-endpoint \
  --endpoint-name my-endpoint \
  --body '{"data": "test"}' \
  --endpoint-url http://localhost:8000 \
  output.json
```

Dashboard (`dashboard/sagemakerrumtime_handlers.go`)
- Invocations table showing operation, endpoint name, input/
output previews, timestamp
- Code snippets for CLI, Go (`github.com/aws/aws-sdk-go-v2/
service/sagemakerruntime`), and Python
- SageMaker Runtime operations rendered in the Docs page via
`SageMakerRuntimeOps`

Tests
- Unit tests for service handler (93.9% coverage)
- Dashboard handler tests (`dashboard/
sagemakerrumtime_handlers_test.go`)
- E2E UI tests (`test/e2e/sagemakerrumtime_test.go`)
- Terraform fixture (`test/terraform/fixtures/sagemakerrumtime/
success.tf`) and integration test
(`TestTerraform_SageMakerRuntime`) verifying `InvokeEndpoint`
via the `sagemakerruntime` SDK

Wiring
- Registered in `cli.go` `getMostRecentServiceProviders()`
- `AWSSDKProvider` interface extended with
`GetSageMakerRuntimeHandler()`
- Added to teststack and E2E test suite
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@agbishop</author>
copilot-autodev: This PR is addressing Issue #458.
</comment_old>
```

<comment_old>
<author>@copilot-pull-request-reviewer</author>
Pull request overview

Adds a new **SageMaker Runtime** service implementation (`sagemaker-runtime`) to the emulator, including a minimal REST handler + in-memory backend, dashboard UI for invocation visibility, and wiring into CLI/teststack/E2E.

Changes:
- Introduces `services/sagemakerrumtime/` with handler/backing store/provider + unit tests.
- Adds a dashboard page + template to display recorded runtime invocations and usage snippets.
- Wires the new service through CLI, dashboard provider/config, teststack, and adds an E2E UI test.

Reviewed changes

Copilot reviewed 13 out of 13 changed files in this pull request and generated 4 comments.

</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>dashboard/sagemakerrumtime_handlers.go:36-45</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
The Go snippet appears to include an unused import (`strings`) and doesn't show the typical pattern for reading the InvokeEndpoint response body. Since snippets are intended to be copy/pasteable, consider removing unused imports and showing reading from the response body (e.g., via io.ReadAll) before logging/printing.

</comment_old>

</comment_thread>

<comment_thread>
<file>services/sagemakerrumtime/handler.go:172-186</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
InvokeEndpointWithResponseStream responses use the AWS event stream binary framing; returning a raw JSON blob here (even with Content-Type set to application/vnd.amazon.eventstream) will cause AWS SDK clients to fail decoding the stream. Consider encoding a valid event stream frame (similar to services/bedrockruntime's encodeEventStreamMsg) and writing that instead of plain JSON bytes.
</comment_old>

</comment_thread>

<comment_thread>
<file>services/sagemakerrumtime/handler.go:65-78</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
RouteMatcher's comment says it matches POST requests, but the matcher currently only checks path prefix and SigV4 service name (no method check). Either update the matcher to also require POST, or adjust the comment so it matches the actual routing behavior.
</comment_old>

</comment_thread>

<comment_thread>
<file>dashboard/doc_handlers.go:154-159</file>
<original_commit>8e6797e</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
docPageData now includes SageMakerRuntimeOps and docIndex populates it, but dashboard/templates/doc.html currently only renders .SageMakerOps (no SageMaker Runtime section). As-is, SageMaker Runtime operations won't appear in the Docs page despite being computed here; consider updating the doc template

to display SageMakerRuntimeOps.
</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@agbishop</author>
@copilot (refinement check 1 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please address any review feedback.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot (refinement check 1 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please addres...

Addressed all four review items in commit `65d1ab3`:

1. **RouteMatcher comment** — updated to accurately describe the matching behavior (path prefix + SigV4 service name check; no POST-only restriction)
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4. **doc.html** — added a SageMaker Runtime section so `SageMakerRuntimeOps` operations are visible on the Docs page
</comment_old>

<comment_old>
<author>@agbishop</author>
copilot-autodev: PR requires manual deployment approval for workflow(s): Pull Request CI. Waiting for a reviewer to approve the environment deployment before proceeding.
</comment_old>

<comment_old>
<author>@agbishop</author>
@copilot (refinement check 2 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please address any review feedback.
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot (refinement check 2 of 2 against issue #458). Please review your implementation against all requirements in the original issue and refine anything that is missing or incomplete. Please addres...

Addressed in commit `dc0b77a`. The gaps found in refinement check 2:

1. **Go snippet package name** — corrected from `sagemakerrumtime` (non-existent) to `sagemakerruntime` (the actual AWS SDK Go v2 package). Also fixed `io.ReadAll(out.Body)` — `InvokeEndpointOutput.Body` is `[]byte`, so it's used directly as `log.Printf("Response: %s\n", out.Body)`.

2. **Dashboard handler tests** (`dashboard/sagemakerrumtime_handlers_test.go`) — added 3 table-driven test cases: renders page header, renders recorded invocation, and shows empty state.

3. **Terraform fixture** (`test/terraform/fixtures/sagemakerrumtime/success.tf`) — created (SDK-direct test, no Terraform-managed resources, same pattern as BedrockRuntime).

4. **Terraform test** (`TestTerraform_SageMakerRuntime`) — added to `terraform_test.go` with `createSageMakerRuntimeClient` helper in `main_test.go` using `github.com/aws/aws-sdk-go-v2/service/sagemakerruntime`. Verifies `InvokeEndpoint` returns a non-empty body.
</comment_old>

<comment_new>
<comment_id>4056391205</comment_id>
<author>@agbishop</author>
copilot-autodev: Merge conflicts were resolved locally by copilot-autodev using `copilot`.
</comment_new>

<comment_new>
<comment_id>4056510313</comment_id>
<author>@agbishop</author>
@copilot (CI-fix 1 of 6). The tests are still failing — please fix them.

Additionally, please fix the following CI failures before pushing:
Failing workflow: Pull Request CI
Failed jobs: terraform (0)

terraform (0) logs: https://productionresultssa14.blob.core.windows.net/actions-results/97f80127-c6d2-404b-8b68-4ab7f5e1ea58/workflow-job-run-a3662ada-56b0-5e2e-8680-77d409a26e6d/logs/job/job-logs.txt?rsct=text%2Fplain&se=2026-03-13T16%3A58%3A05Z&sig=ul%2FxbZ4trJeGthHYXWSCwKwNAJPr2yYGBj3BSSl8MMU%3D&ske=2026-03-13T18%3A58%3A00Z&skoid=ca7593d4-ee42-46cd-af88-8b886a2f84eb&sks=b&skt=2026-03-13T14%3A58%3A00Z&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skv=2025-11-05&sp=r&spr=https&sr=b&st=2026-03-13T16%3A48%3A00Z&sv=2025-11-05
</comment_new>
</pr_comments>

</comments>

The last **6** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement><rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
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* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.

```
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>




A **CI/Build Failure** has been mentioned in the problem
statement, make sure to follow your instructions for addressing
CI/Build Failures.
```